

# How to Get an LLM API Key and Use It with Ouvoca (Complete Guide)

Applies to Ouvoca v3.14+ Target audience: non-technical users Estimated time: 5–10 minutes

## 1. Why Do I Need an API Key?

Ouvoca's "AI Conversational CRUD" feature (query/add/edit/delete by talking) is powered by a **Large Language Model (LLM)**. LLMs run in the cloud — to call one, you must first **sign up for an account and apply for an API Key** (like a key).

### What If I Don't Have an API Key?

Ouvoca is **designed to work without one**:

| Feature                                                     | API Key Required? |
|-------------------------------------------------------------|-------------------|
| Login / account management                                  | ✗ No              |
| Inventory / Purchase / Sales / Production (mouse-click ops) | ✗ No              |
| File upload / report download                               | ✗ No              |
| Load demo data                                              | ✗ No              |
| War-room real-time dashboard                                | ✗ No              |
| AI Assistant chat (CRUD)                                    | ✓ Yes             |

In other words: **You can use Ouvoca as a traditional ERP without applying**, but you lose the "talk to it" headline feature.

## 2.3 LLM Providers Compared (Pick One)

| Provider                  | Price                                 | Free Tier              | Chinese Quality | Recommended For               |
|---------------------------|---------------------------------------|------------------------|-----------------|-------------------------------|
| cn DeepSeek (recommended) | \$0.14 / 1M input tokens (dirt cheap) | Small credit on signup | ★★★★★           | Most users: best value        |
| us OpenAI (GPT-4o-mini)   | \$0.15 / 1M                           | \$5 trial (90 days)    | ★★★★            | Already have OpenAI account   |
| us Anthropic Claude       | \$0.25 / 1M (haiku)                   | \$5 on signup          | ★★★★★           | Better reasoning, long-form   |
| 🏠 Ollama (offline)        | \$0 (runs locally)                    | Unlimited              | ★★★             | Have GPU + want fully offline |

### Our Recommendation → DeepSeek

Reasons:

1. 100× cheaper than OpenAI
2. Best Chinese understanding (especially Traditional Chinese)
3. OpenAI-compatible API format — easy to switch providers later
4. A typical small factory's monthly bill is NT\$10–100 (essentially free)

If you're worried about a Chinese company storing your data → pick OpenAI or Claude (data stays in the US).

## 3. Apply for a DeepSeek API Key (Recommended, 5 min)

### Step 1 · Sign Up

1. Open [https://platform.deepseek.com/sign\\_up](https://platform.deepseek.com/sign_up)
2. Register with **email** or **phone number** (Gmail works best)
3. Verify email, set password, land on home page

### Step 2 · Get API Key

1. After login, click "API Keys" in left sidebar
2. Click blue "Create new API Key" button (top right)
3. Name the key (e.g., `ouvoca-2026`), click Create

4. Copy the `sk-xxxxxxxxxxxxxxxxxxxxxxxxxxxx` shown immediately (shown only once!)

⚠️ Forgot to copy → delete and create a new one. Don't try to find it in the page.

## Step 3 · Top Up (Optional)

The free signup credit is limited (about NT\$15 / ~200 chats).

To continue using:

1. Click "Top Up" in left sidebar
2. Minimum top-up is USD\$2 (~300 chats)
3. Credit card / PayPal / Alipay / WeChat Pay all accepted

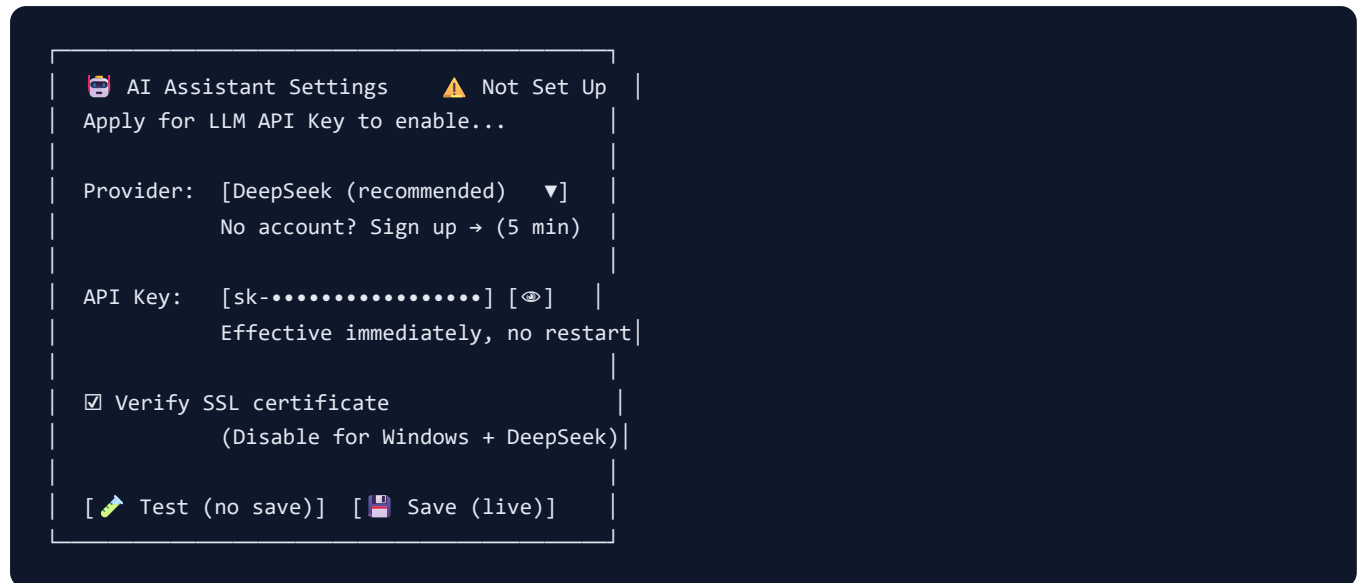
💡 **Saving tip:** NT\$30/month is plenty for a small factory's ERP chat volume. Try the free tier first to see how quickly you burn through it.

## 4. Give the Key to Ouvoca (3 Methods)

### Method A: Settings Page (recommended, easiest)

1. Login to Ouvoca (<http://localhost:5173>)
2. Click "⚙️ Settings" in left sidebar
3. In the top "🤖 AI Assistant Settings" section:
  - **Provider** → pick "DeepSeek" from dropdown
  - **API Key** field → paste your `sk-xxxxx...`
  - Click "🔧 Test Connection (no save)" → see "✅ DeepSeek connection successful"
  - Click "💾 Save (effective immediately)"
4. Done! No restart, no re-login. Go to AI Assistant page and start talking.

## Screenshot Mockup (ASCII art)



## Method B: Edit backend/.env (for developers)



Save, then:

- Docker mode: `docker compose restart backend`
- Dev mode: run `stop_dev.bat` → `start_dev.bat`

## Method C: Environment Variables (advanced / production)



## 5. Windows + DeepSeek Common Issue: SSL Certificate Error

### Symptom

Click "Test Connection" and see:

```
✖ Connection failed: SSLCertVerificationError
```

### Cause

Windows' CA cert store hasn't updated with DeepSeek's root certificate. Common issue with Windows + DeepSeek combo — not an Ouvoca bug.

### Fix (fastest)

In Ouvoca Settings, **uncheck "Verify SSL certificate"** → retry test. This just makes Python skip SSL verification — doesn't affect your security (key is still sent over TLS-encrypted connection).

Cleaner fix: upgrade Python `certifi` package — not necessary for non-developers.

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## 6. Other Provider Steps

### OpenAI (GPT-4o-mini)

1. Sign up: <https://platform.openai.com/signup> (credit card required)
2. Get Key: left API Keys → **Create new secret key** (shown once)
3. In Ouvoca Settings: pick OpenAI, paste key, save

### Anthropic Claude

1. Sign up: <https://console.anthropic.com/>
2. Go to **Workbench** → **Settings** → **API Keys** → **Create Key**
3. In Ouvoca Settings: pick Anthropic, paste key, save

### Ollama (local offline)

1. Download: <https://ollama.com/download> → install
  2. Terminal: `ollama pull llama3.2` (~2GB model)
  3. In Ouvoca Settings: pick Ollama, leave key empty, save
  4. Confirm `ollama serve` is running locally
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## 7. Security Reminders

### ✓ Things You Should Do

- API Key = password; leaks let others use your quota
- Don't paste keys into chat / Slack / GitHub commits
- Rotate keys regularly (every 3-6 months)
- If you suspect a leak → revoke at provider + create new

### ✓ How Ouvoca Handles Your Key

- Stored in `backend/.env` on your local machine (plaintext file)
- **Never** uploaded to any server
- **Never** git-committed ( `.gitignore` excludes `.env` )
- Pre-commit hook auto-scans `sk-` patterns and blocks accidents

### ✓ Where Conversation Data Goes

When you chat with the AI Assistant:

- Your **typed text** is sent to the LLM provider (DeepSeek/OpenAI/Anthropic)
- **Business DB data is not sent** (unless the AI calls a tool that queries DB and includes results in subsequent context)
- Provider retention policies differ — check their privacy terms

To keep data fully on-premise → use **Ollama offline mode** (Method D).

## 8. Cost Estimates (Small Factory Reality)

| Scenario                          | Usage       | DeepSeek/mo | OpenAI/mo |
|-----------------------------------|-------------|-------------|-----------|
| Boss asks 5× daily                | 150/month   | NT\$3       | NT\$45    |
| Sales rep uses 10× daily          | 1500/month  | NT\$30      | NT\$450   |
| Whole 20-person company heavy use | 20000/month | NT\$400     | NT\$6,000 |

💡 Ouvoca is token-efficient: each chat usually < 2000 tokens.

## 9. Troubleshooting

| Symptom                  | Fix                                                            |
|--------------------------|----------------------------------------------------------------|
| "Connection refused"     | Check network / proxy settings                                 |
| "Invalid API key"        | Re-confirm key has no missing chars / extra spaces             |
| "Rate limit"             | Too many requests, wait minutes or upgrade plan                |
| "Insufficient balance"   | DeepSeek balance empty, go Top Up                              |
| "Model not found"        | Provider changed model name; update in Settings                |
| Windows SSL error        | Uncheck "Verify SSL certificate"                               |
| Settings won't save      | Confirm you're admin (need <code>system.config.update</code> ) |
| Key change has no effect | Save is instant; if not, hard refresh (Ctrl+F5)                |

## 10. Related Documentation

- [USER\\_MANUAL\\_EN.md](#) — Full user manual
- [PRODUCT\\_OVERVIEW\\_EN.md](#) — Product overview for buyers
- [INSTALLATION\\_EN.md](#) — Installation guide
- [LICENSE-SMALL-BUSINESS.md](#) — Free tier (≤20 users)

Chinese version: [HOW\\_TO\\_GET\\_LLM\\_API\\_KEY\\_ZH.md](#)